

Classical Field Theory

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The Euler-Lagrange Equations

The action S is defined as

$$S = \int dt L = \int d^4x \mathcal{L}, \quad (1.1)$$

where L is the Lagrangian of the system, and $\mathcal{L}$ is the Lagrangian density. Say we have a Lagrangian $\mathcal{L}[\phi, \partial_\mu \phi]$ is only a functional of ϕ and $\partial_\mu \phi$. Then,

$$\begin{aligned} \delta S &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta (\partial_\mu \phi) \right] \\ &= \int d^4x \left\{ \left(\frac{\partial \mathcal{L}}{\partial \phi} - \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \delta \phi + \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta (\phi) \right] \right\}. \end{aligned} \quad (1.2)$$

The last term is a total integral and the boundary condition will always force it to vanish. Imposing the principal of least action, $\frac{\delta S}{\delta \phi} = 0$. Hence, this implies

$$\frac{\partial \mathcal{L}}{\partial \phi} - \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = 0. \quad (1.3)$$

Noether's Theorem

A Lagrangian may be invariant under some transformation $\phi \rightarrow \phi + \delta \phi$, which is called a symmetry. When there is such a symmetry that depends on some parameter α that can be taken small (a continuous symmetry), we find, similar to (1.2)

$$0 = \frac{\delta \mathcal{L}}{\delta \alpha} = \sum_n \left(\frac{\partial \mathcal{L}}{\partial \phi_n} - \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \right) \frac{\delta \phi_n}{\delta \alpha} + \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha} \right]. \quad (2.1)$$

When the equation of motion is satisfied, then (2.1) simplifies to $\partial_\mu J^\mu = 0$ with

$$J^\mu = \sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha}. \quad (2.2)$$

This is known as the Noether current. Such a vector field is also called a conserved current, since the total charge Q , defined as

$$Q = \int d^3x J_0, \quad (2.3)$$

satisfies

$$\partial_t Q = \int d^3x \partial_t J_0 = \int d^3x \vec{\nabla} \cdot \vec{J} = \int d^3x \vec{S} \cdot \vec{J} = 0. \quad (2.4)$$

Energy-Momentum Tensor

In this section, we will investigate the conserved current corresponding to space-time translation invariance, known as the energy-momentum tensor, $\mathcal{T}_{\mu\nu}$. Now, let us consider the transformation $y^\mu \rightarrow y^\mu - \xi^\mu$ with ξ^μ to be a constant four-vector. The scalar fields then would transform as $\phi(x) \rightarrow \phi(x + \xi)$. For infinitesimal $\delta\xi^\mu$

$$\phi(x) \rightarrow \phi(x + \delta\xi) = \phi(x) + \delta\xi^\mu \partial_\mu \phi(x) + \dots \quad (2.5)$$

This implies

$$\frac{\delta \phi(x)}{\delta \xi^\mu} = \partial_\mu \phi(x). \quad (2.6)$$

As the Lagrangian is also a function of x , we have

$$\frac{\delta \mathcal{L}}{\delta \xi^\mu} = \partial_\mu \mathcal{L}. \quad (2.7)$$

Since this is a total derivative, $\delta S = \int d^4x \delta \mathcal{L} = \delta \xi^\mu \int d^4x \partial_\mu \mathcal{L} = 0$, which is why we sometimes say this is a symmetry of the action, not the Lagrangian. Similarly to before, we have

$$\frac{\delta \mathcal{L}[\phi_n, \partial_\mu \phi_n]}{\delta \xi^\nu} = \partial_\mu \sum_n \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \xi^\nu} \right] = \partial_\mu \sum_n \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n(x) \right]. \quad (2.8)$$

Thus,

$$\begin{aligned} & \partial_\nu \mathcal{L} - \partial_\mu \sum_n \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n(x) \right] = 0 \\ \implies & \partial_\mu \left(\sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} \mathcal{L} \right) = 0. \end{aligned} \quad (2.9)$$

We obtained a conserved current with four components, being energy and momentum. Therefore, we define the energy-momentum tensor $\mathcal{T}_{\mu\nu}$ to be

$$\mathcal{T}_{\mu\nu} = \sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} \mathcal{L} \quad (2.10)$$

with $\partial_\mu \mathcal{T}_{\mu\nu} = 0$.

Green's Functions

Now, we will introduce a very powerful tool in field theory, the Green's functions. Suppose we have a very general differential equation formed from a linear operator $\hat{L}$

$$\hat{L}x(t) = f(t). \quad (3.1)$$

The Green's function $G(t, u)$ (or $\Pi(t, u)$) of the equation is defined as

$$\hat{L}G(t, u) = \delta(t - u). \quad (3.2)$$

Let us consider a familiar example from mechanics. We will take an oscillator of mass m and spring constant K evolving under the influence of a time-dependent force $f(t)$. The equation of motion for this is

$$m \frac{d^2}{dt^2} x(t) + Kx(t) = f(t), \quad (3.3)$$

so here the linear operator is $\hat{L} = m \frac{d^2}{dt^2} + K$. We can build up a force function $f(t)$ by adding together lots of delta functions.

$$f(t) = \int_0^\infty du f(u) \delta(t - u), \quad (3.4)$$

where u is a dummy variable which allows us to break $f(t)$ into deltas. This is, of course, merely an example of superposition and is allowed because our differential equation is linear.

Instead of hitting the problem head-on, we sidestep and solve Eq. (16.3) for just one of the delta functions. We'll call this solution the Green's function $G(t, u)$:

$$\left[m \frac{d^2}{dt^2} + K \right] G(t, u) = \delta(t - u). \quad (3.5)$$

The complete solution to Eq. (16.3) is then given by the integral of the Green's function, weighted by the force function $f(u)$:

$$x(t) = \int_0^\infty du G(t, u) f(u). \quad (3.6)$$

The solution is therefore simply a sum of the responses of the system in x to the stimuli $f(u)$.

Why is the solution to Eq. (16.3) given by Eq. (16.6)? Start by acting on Eq. (16.6) with the operator $\hat{L}$ (which passes through the integral sign on the right-hand side),

$$\begin{aligned}\hat{L} x(t) &= \int du \hat{L} G(t, u) f(u) \\ &= \int du \delta(t - u) f(u) \\ &= f(t)\end{aligned}\tag{3.7}$$

which is where we started! This implies we can solve an inhomogeneous differential equation by finding the Green's function $G(t, u)$ and then integrating over $f(u)$ to get the solution $x(t)$.